Particle NeRD

Learn particle motion.
Reconstruct body reactions analytically.

Replacing the expensive MPM update while retaining interaction with rigid ground and moving bodies.

Latest experiment progress →

LOCAL CODE

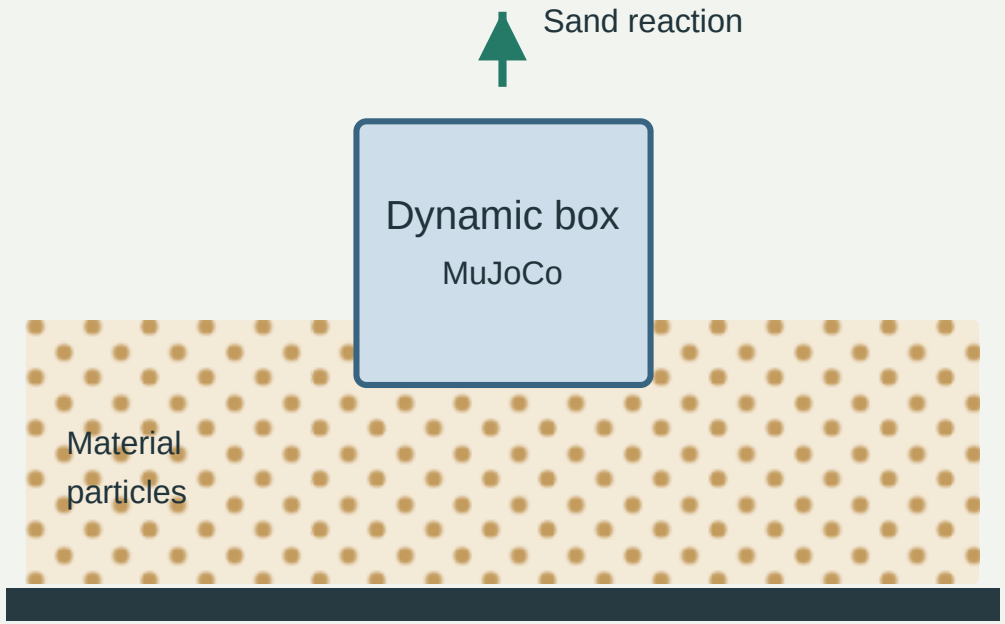
A sand bed, rigid ground, and one dynamic body

MuJoCo advances the rigid body.
MPM advances the sand and supplies contact impulses.

Keep Rigid-body dynamics, rigid-body contacts, and Newton’s coupling interface.

Replace The expensive material update with learned motion and analytical reaction reconstruction.

Dynamic collection is being implemented with MJWarp–MPM coupling. Earlier results use a prescribed box; see the progress page.



Fixed rigid ground beneath the sand

Particle motion and body reaction must remain consistent.

Source: Newton coupled MPM–MuJoCo example. Illustration is schematic.

LOCAL CODE

Common geometry; different motion constraints

Newton converts supported shapes into collider meshes and rasterizes contact information onto the MPM grid.

COLLIDER	GEOMETRY AND CONTACT PROPERTIES	MOTION	REACTION DESTINATION
Rigid ground	Mesh, distance, normal, friction	Fixed	Fixed environment
Prescribed tool	Same representation	Externally supplied	Measured reaction; pose remains prescribed
Dynamic body	Same representation	Rigid-body solver	Rigid-body solver

Implication for NeRD Use shared collider features. Preserve each collider’s identity, motion, and material properties; separate “ground” and “shovel” networks are not required.

Sources: shape conversion · collider queries · rasterization and coupling

SLIDE DECK

The cable model learns motion corrections

01

Collect

Record cable trajectories from VBD under randomized manipulation and contact.

Segment position, orientation, and motion history.

02

Train

Start from a recorded state. Predict corrections to motion under inertia and gravity.

Supervise one-step corrections using the next VBD frame.

03

Infer

Apply predicted corrections, enforce constraints, and advance from the model's own state.

VBD is absent from the cable inference loop.

```
baseline motion + learned correction → constraint enforcement → next state
```

Source: cable representation and training. The deck does not specify the exact baseline integration formula.

CABLE SLIDES 23–24

The cable reconstructs forces after predicting motion

Motion history
Three segment poses → accelerations

→

Known physics
Mass, inertia, gravity and rod bending law

→

Chain solve
Joint and contact force / torque balance

→

Body reaction
Sum opposite contact forces and torques

What makes it tractable

Fixed chain connectivity creates a block-tridiagonal system. Geometry restricts where external contact forces may act.

What the network supplies

Motion only. The reconstruction uses physical equations and softly enforced rod moments; no force network is required.

Transferable principle

Learn the expensive motion update; reconstruct reactions with known mechanics.

Source: force reconstruction · chain implementation. The cable implementation applies delayed, capped body reactions.

METHOD DISTINCTION

Constraint solving and force reconstruction are related

XPBD

Solve for constrained motion

Correct predicted positions using physical constraints. Constraint multipliers provide force estimates.

Prediction → constraint solve → motion + forces

CABLE NeRD

Reconstruct forces from motion

Use predicted motion, contact geometry, and material laws to solve for consistent reactions.

Learned motion → balance equations → forces

A collision push-out is insufficient

If NeRD already predicts a supported, nonpenetrating state, the final push-out can be zero while the physical support force is nonzero.

Sources: XPBD, Macklin et al. (2016) · cable slide 23

PAPER + RELEASED CODE

L-GBND separates material, tool, and static nodes

NODE TYPE IN RELEASED CODE	POSITION UPDATE	ROLE
Terrain / type 0	Current position + learned displacement	Deforming material
Tool / type 1	Current position + prescribed action	Moving contact geometry
Wall / type 2	Unchanged	Static environment

Useful precedent. A graph can represent material–tool interaction with distinct node roles.

Missing for our task. The released position predictor does not return a reaction wrench to a freely moving body.

The paper’s region-boundary normals represent omitted stationary terrain. They do not specify an explicit rigid-ground collision solve.

Sources: L-GBND, Sections III-A–D · released node updates

DESIGN

One material model, explicit collider information

PARTICLE STATE

Motion and neighborhood

Positions, velocities, masses, and relative positions and velocities of nearby material samples.

Record APIC velocity gradients; test whether they are needed as model state.

CONTACT INPUTS

Geometry and motion

Distance, surface normal, relative surface velocity, friction, and collider identity.

Include both ground and body features where both are relevant.

MODEL OUTPUT

Particle motion

A learned displacement correction to an explicit gravity-and-velocity baseline.

No learned body-reaction output. No MPM solve at inference.

Update every material particle

Use a small complete sand bed with local graph connections. No region selector; keep material, timestep, and resolution fixed initially.

Proposal informed by cable NeRD and Newton's APIC state and collider inputs.

OPEN

What replaces the cable’s chain equations?

```
mass × particle acceleration
= gravity force + internal material force + collider contact forces
```

Known from motion and geometry

Acceleration, mass, gravity, potential contacts, contact normals, and collider motion.

Unknown

Internal stress transmission and the separate contact reactions on ground and each body.

Candidate reconstruction

Solve local balance on a coarse grid or particle neighborhood graph, constrained by material response and admissible contact forces.

Compression, friction, and material assumptions must be defined. Neighborhood edges are not cable joints.

An initial particle-graph reconstruction has now been tested; see slides 16–18. A suitable granular material model remains an open question.

BEFORE NETWORK TRAINING

Test reconstruction using MPM’s own motion

MPM trajectories

Motion history + geometry
+ known material parameters

→

Analytical reconstruction

Estimate body reaction
without MPM force labels

→

Reference comparison

Compare with MPM impulses
and body torques

EXPERIMENT	WHAT IT ESTABLISHES
Sand at rest on rigid ground	Recovered ground support balances sand weight.
Box supported by sand, clear of rigid ground	Recovered sand force supports the box’s weight.
Box pressing into and sliding through sand	Reconstruction captures changing normal load, friction, and torque.

Use an MPM iteration cap of 25–50. Check static force balance and stable motion; increasing the iteration cap is not a prerequisite.

PLANNED DATASET

Record motion, contacts, and validation references

Generate varied MPM episodes

- Settling and static support.
- Vertical pressing and release.
- Lateral pushing and sliding.
- Different starting depths and body orientations.

Use prescribed trajectories for motion-model experiments and dynamic-body episodes for coupling evaluation.

Save each timestep

- Particle position, velocity, mass, and APIC gradient.
- Body pose, velocity, and contact material parameters.
- Material configuration, timestep, and resolution.
- MPM body impulses and torques for validation.

Split by episode

Reserve complete motions and initial conditions for validation and testing. Adjacent frames from one episode must not cross dataset splits.

MPM baseline: 25 iterations maximum, tolerance 1e-4, following the G1 configuration. Reaction labels are for validation only.

IMPLEMENTATION

Train on MPM transitions; select by rollout accuracy

Recorded state at t
Particle state
+ collider inputs

→

Baseline motion
Current velocity
+ gravity

→

NeRD correction
Particle interactions
+ contact response

→

MPM state at t + 1
Supervised
motion target

```
target displacement correction = MPM next position − baseline next position
```

Initial training recipe. One-step supervision with independent position and velocity perturbations; test noise scale and loss choice.

Checkpoint selection. Repeated model rollouts with identical collider trajectories. Measure drift, penetration, and material redistribution.

Proposed modules: `particle_model.py` · `train_motion.py`. The cable recipe motivates these choices; particle stability remains to be tested.

IMPLEMENTATION

Close the loop with an analytical reaction step

MuJoCo rigid-body update

Pose and velocity under applied body wrenches

→

Particle NeRD update

Baseline + learned correction + geometric constraints

Newton coupling interface

Per-body force and torque for the next coupled step

←

Analytical reconstruction

Particle motion + material model + contact geometry

↑ Apply reconstructed reaction

↓ Reconstruct after the particle update

Retain collider identity throughout

Sum each contact reaction onto the correct body. Compute torque about that body's center of mass; keep fixed-ground reactions separate.

Proposed adapter: `solver_particle_nerd.py`. Integration target: Newton's proxy-wrench interface.

RESULTS ON SLIDES 15–18

Implementation milestones

MILESTONE	TASK	SUCCESS CRITERIA
1 · Data collection	Record particle motion and body reactions using an MPM iteration cap of 25–50.	Particle and body trajectories remain stable. Support forces balance weight in static tests.
2 · Force reconstruction	First check whether body reactions are uniquely determined. Then reconstruct forces using explicit material and contact relationships.	Estimated reactions agree with MPM and take less time to compute than the MPM update.
3 · Motion prediction	Train a graph neural network to predict particle motion from MPM trajectories.	Predicted trajectories match MPM on test episodes over multiple timesteps.
4 · Coupled simulation	Use NeRD and reconstructed reactions in Newton to simulate particles and moving bodies.	Body motion and sand deformation remain close to MPM without running the MPM solver.
5 · Larger scenes	Increase the number of particles while updating every particle at each timestep.	The complete simulation runs faster than MPM while maintaining the required accuracy.

Define acceptable motion and force errors before evaluation. Measure runtime for the complete simulation, including contact queries and force reconstruction.

COLLECTION CHECKS PASSED

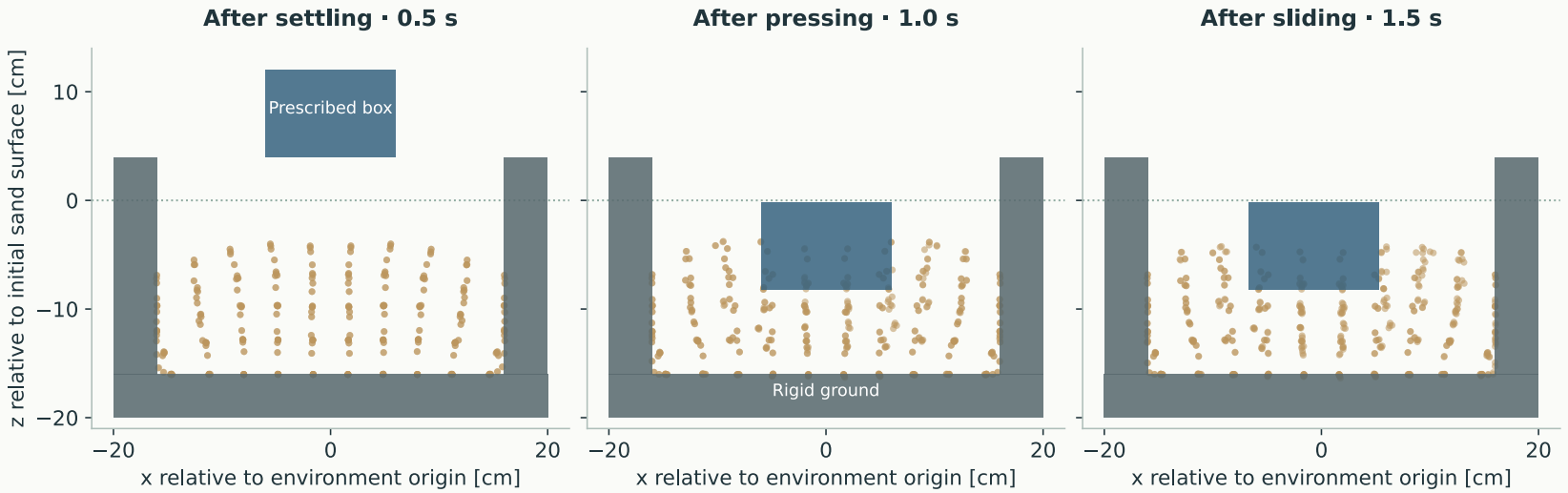
Reference data collected from parallel MPM simulations

2
parallel environments

500
particles per environment

4 × 2 s
recorded episodes

1,600
recorded transitions



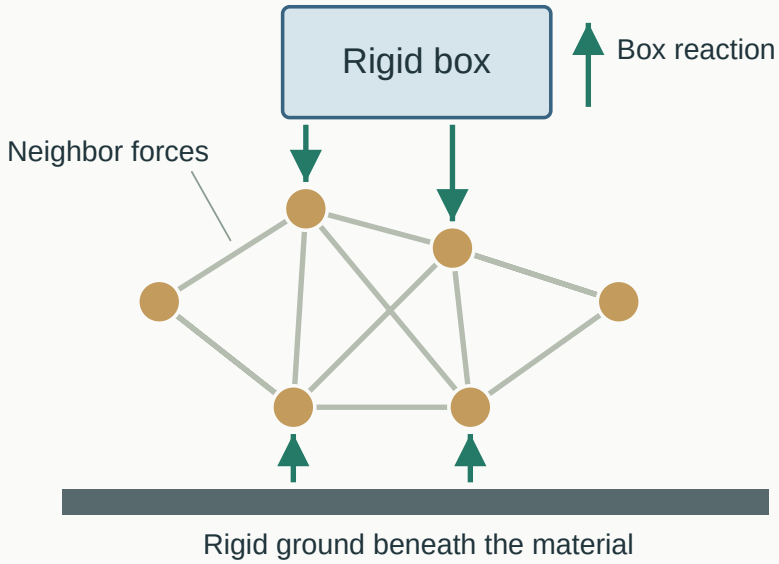
Actual recorded positions, projected onto the x–z plane. Front and rear walls are omitted from the illustration.

Verified: recording alignment, episode resets, particle retention, settling, and total linear momentum balance.

25 MPM iterations maximum · tolerance 1e-4 · timestep 5 ms · momentum residual 0.00084% of the reference scale.
Small validation dataset; no model training. Recorded data retained locally · Implementation and run commands

EXPERIMENTAL ASSUMPTION

Estimate body reactions from recorded particle motion



Schematic of the reconstruction assumption; not an MPM force network.

1. Measure acceleration
Use the change in each particle's velocity.

2. Solve particle force balance
Allow compressive forces between neighbors and friction-limited contact forces.

3. Sum reactions on each body
Apply equal-and-opposite contact forces and their moments about the body center.

Key assumption

A small force penalty selects one solution when motion admits several force distributions. This does not reproduce Newton's granular material law.

Inputs: MPM particle motion, mass, gravity, and rigid-box geometry. MPM reaction forces are withheld until evaluation.
No neural model is used. All particles participate; ground and moving boxes use the same contact representation.

SUCCESS CRITERIA NOT MET

The current reconstruction misses the box reaction



One recorded episode. Positive force points upward.
Reconstruction markers show only the four evaluated times.

38.5%

relative body-force error
across 16 sampled frames

79.9 N

moving-box force RMSE
across those 16 frames

7.0 s

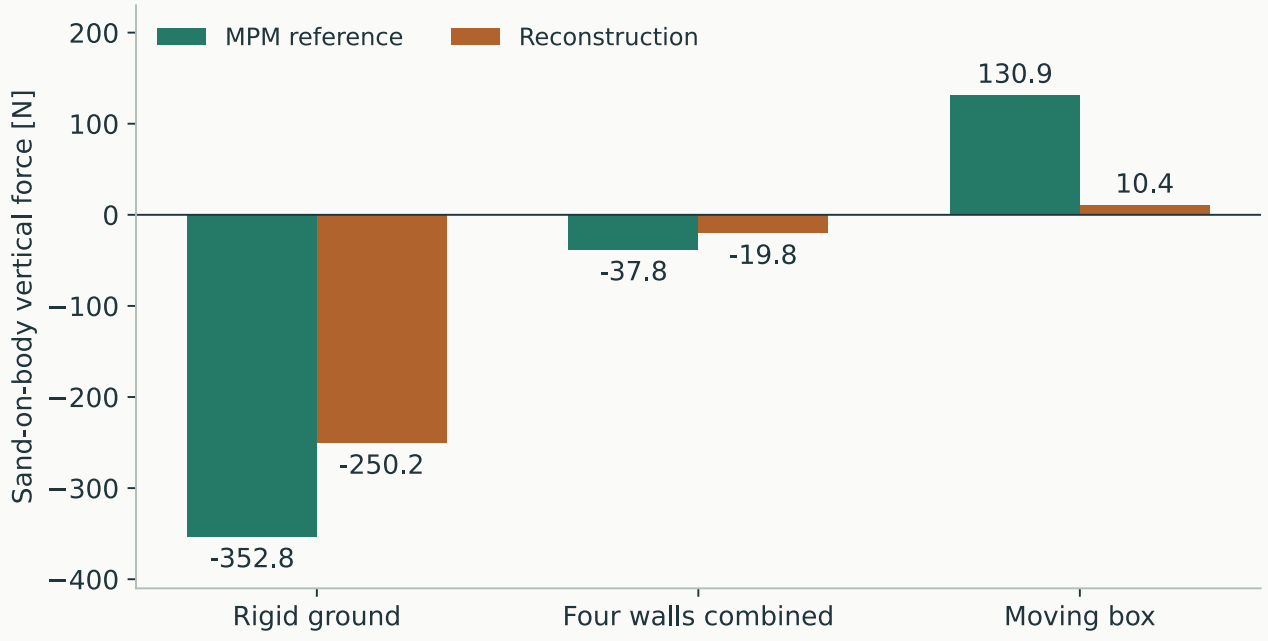
median CPU reconstruction time
per sampled frame

16 / 16 solves reached the iteration limit.
These are provisional estimates.

Relative error = norm of all estimated-minus-reference body forces / norm of all reference body forces.
Four frames per episode, six bodies per frame; force error includes all three components. Full evaluation results (local file).

MECHANICS STILL UNRESOLVED

Correct total force can hide incorrect body forces



Episode 0 at 1.0 s: force exerted by sand on each body.
Positive is upward; negative is downward.

Total vertical force on all bodies

−259.65 N MPM

−259.65 N reconstruction

The totals agree, but the estimated upward force on the moving box is much too small.

Implication: satisfying particle force balance does not validate the force on each body.

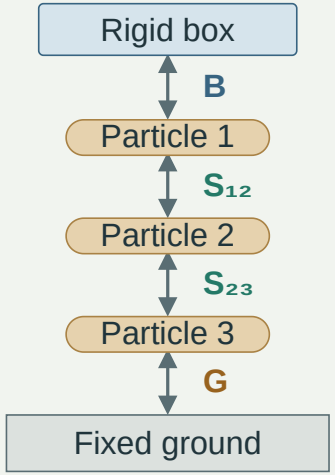
Next: check whether the full particle equations uniquely determine body reactions, before choosing a force-selection rule.

Optimization error and the material-force assumption are both unresolved. A separate one-frame diagnostic converged under adjusted tolerances but retained 44.8% body-force error; it is not a full reevaluation.

ONE-DIMENSIONAL EXAMPLE

Equations link the rigid body, particles, and ground

Upward is positive. Each contact force appears with opposite signs on the two objects it connects.



B, S_{12}, S_{23}, G: compressive force magnitudes at successive contacts.

Rigid box	$m_b\, a_b = U + B - W_b$
Particle 1	$m_1\, a_1 = S_{12} - B - W_1$
Particle 2	$m_2\, a_2 = S_{23} - S_{12} - W_2$
Particle 3	$m_3\, a_3 = G - S_{23} - W_3$
Fixed ground	$0 = R - G - W_g$

m = mass; a = vertical acceleration; W = weight (m g).
U: external actuator force on the box. **R:** external support force on the ground. Both are signed upward.

Add all five equations: $m_b\, a_b + m_1\, a_1 + m_2\, a_2 + m_3\, a_3 = U + R - \Sigma_i\, m_i\, g$

Contact forces cancel in the sum. **The actuator force U and support force R remain.**
With known U and measured box acceleration, B follows from the box equation. With prescribed box motion and unknown U, that equation introduces another unknown.

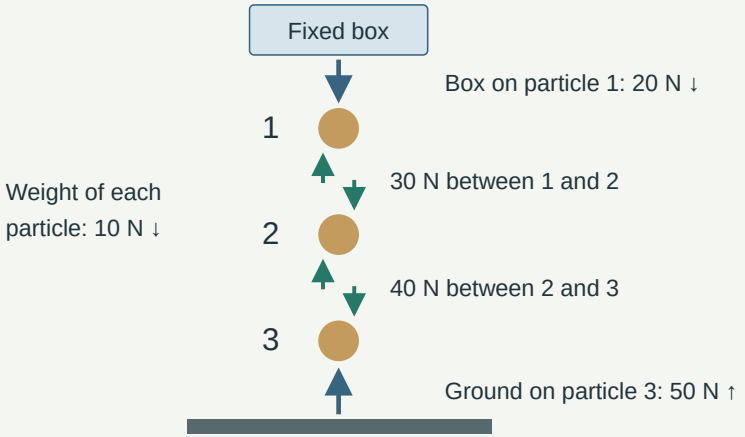
Sum over the box, particles 1–3, and supported ground slab. Vertical translation only; zero ground acceleration does not imply zero contact force.

EXACT FORCE-BALANCE EXAMPLE

Different contact forces can produce identical motion

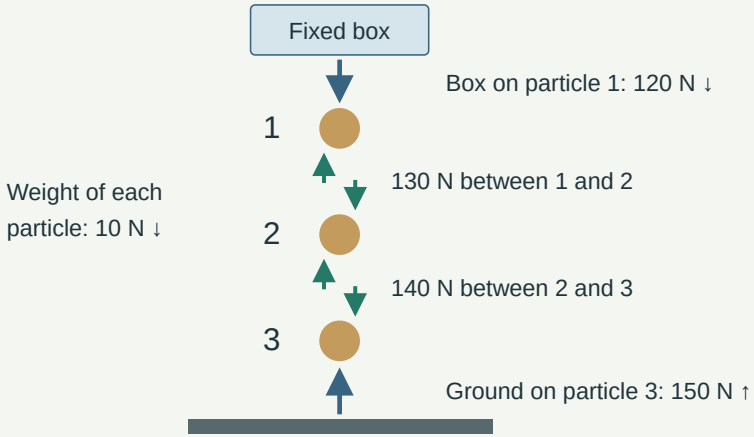
Three stationary particles. Only particle 1 touches the box; only particle 3 touches the ground.

A · Lower compression



Particle 1	30 − 20 − 10 = 0 N
Particle 2	40 − 30 − 10 = 0 N
Particle 3	50 − 40 − 10 = 0 N

B · Add 100 N along the entire chain



Particle 1	130 − 120 − 10 = 0 N
Particle 2	140 − 130 − 10 = 0 N
Particle 3	150 − 140 − 10 = 0 N

Every particle equation is satisfied in both cases. Adding the equations gives ground force − box force = 30 N in either case.

Upward force is positive. Equal-and-opposite internal forces cancel. Identical geometry and zero acceleration do not select A or B.

This example uses force balance and compressive contact only; a material law or a known boundary load could provide the missing information.

NOT YET COMPLETED

Check whether the equations determine each body reaction

1. Assemble all balances

Balance each particle's forces. Internal forces have opposite signs on neighbors. Include contact forces only where contact occurs.

→

2. Look for force changes

Can forces change while every balance remains satisfied and contact constraints remain valid?

→

3. Check body reactions

If those changes alter a body's net force or torque, motion alone has not determined that reaction.

Some ambiguity is acceptable

Internal forces may be nonunique while every body reaction is unchanged. We only need the body reactions to be determined.

Otherwise, identify missing information

Examples: a material force–deformation relationship, material history, or a known applied boundary load. Test whether it resolves the ambiguity.

Correction to the work sequence

This analysis should have preceded the minimum-force implementation. That implementation is an exploratory baseline, not a validated reconstruction method.

Inference requirement: NeRD particle motion + nonlearned reaction reconstruction. No MPM solve at inference.
MPM is used only to generate reference data and evaluate results.

IMPLEMENTED BASELINE

The current estimate omits most material properties

ASSUMPTION	WHAT THE IMPLEMENTATION USES
Particle–particle forces	Particles within 6.5 cm transmit equal-and-opposite compressive forces along their connecting line. No separate tangential force.
Particle–body forces	Particles within 4 cm of a body surface are contact candidates. Forces obey a four-sided approximation of the friction cone.
Choice among possible solutions	A quadratic penalty favors small force coefficients. It supplies no force–deformation relationship.

Current objective minimize: force-balance error² + 0.0001 × Σ force coefficient²

Parameters used directly

Particle mass and particle–body friction. Density affects the recorded particle mass.

Material properties not used directly

Young’s modulus, Poisson’s ratio, internal material friction, plasticity, and material history.

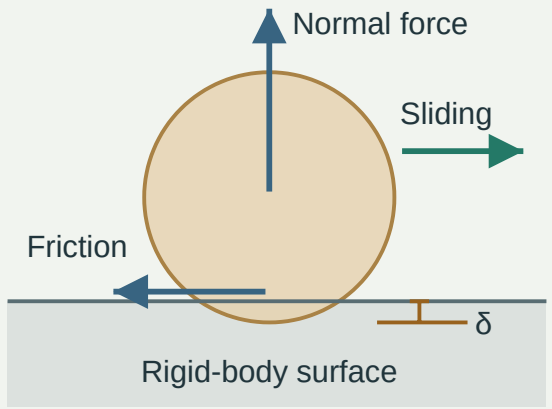
With identical motion, masses, geometry, and body friction, changing an omitted material parameter does not change the estimate. It can affect the result indirectly by changing the motion.

Source: current reconstruction implementation. Small force-balance error does not establish accurate forces on individual bodies.

SOURCE OF ADDITIONAL EQUATIONS

Relate each contact force to deformation and motion

A simple noncohesive DEM contact law supplies force values from overlap, relative velocity, and contact history.



δ = particle–surface overlap

Illustration of a DEM grain.
MPM material points are not physical grains.

Normal force magnitude · active contact

$$F_n = \max(0,\ k_n\,\delta + c_n\,v_{close})$$

Elastic compression + damping; closing speed is positive.

Tangential force vector · before sliding

$$F_{t,trial} = -k_t\,\xi - c_t\,v_t$$

ξ accumulates relative tangential displacement during contact.

Sliding limit

$$\|F_t\| \le \mu\,F_n$$

Cap the trial force and update history when sliding occurs.

Additional equations: force is related to contact deformation and history, rather than selected only by a small-force preference. Separated particles carry no contact force.

k: contact stiffness; c: damping coefficient; μ: contact friction. Ground and moving bodies use the same law with their own geometry and surface velocity.
Source: LAMMPS granular contact laws and particle–wall contacts. The equations show a linear spring–damper model; Hertz–Mindlin is an alternative.

SOURCE REVIEWED

Warp provides a simple DEM contact-force kernel

Linear spring–damper contact, velocity-based friction, and short-range cohesion. No tangential history.

Normal force on particle i

$$F_n = -s\,n$$

$s = k_n\,c + k_d \min(v_n, 0)$

Tangential force magnitude

$$\|F_t\| = \min(k_f\,\|v_t\|, \mu\,|s|)$$

Direction opposes tangential relative velocity.

For particle pairs: c = center distance − 2 × radius; n points toward particle i; v_n = n · relative velocity.

Cohesion	Positive gaps can produce attraction within a cutoff.
Static friction	Zero tangential speed gives zero friction; no stored tangential displacement.
Geometry	Equal-radius particles and a fixed plane; no arbitrary moving-body contacts.

Proposed reuse: hash-grid neighbors and contact-force evaluation. Add rigid-body surface velocity and reaction accumulation. NeRD supplies motion; do not use the example's particle integration.

Source: NVIDIA Warp · example_dem.py. This is a standalone Warp example, not Newton's implicit MPM solver.

PRELIMINARY EVALUATION COMPLETE

Test a contact law on the recorded particle motion

Evaluated objective minimize: force-balance error² + λ × contact-law error²

The second term favors forces predicted by the contact law. λ sets its weight relative to motion consistency.

1. Define effective contacts

Specify reference spacing or effective radii, contact stiffness, damping, and friction. Do not equate MPM spacing with grain overlap.

→

2. Compare contact laws

Evaluate velocity-only friction and a history-based law. Use particle and body velocities; maintain history through each frame.

→

3. Evaluate body reactions

Compare with MPM reference forces on each body, including densely sampled impact intervals and small position perturbations.

Main modeling question

What effective deformation should determine force? MPM particles sample a continuum; their radius is not automatically a grain radius.

Success criteria

Lower body-force error with small balance residual, stable estimates under position errors, and acceptable reconstruction time.

A penetration-only law gives zero support force at zero overlap and zero relative speed. Reference geometry and contact compliance must therefore be defined and tested.

Inference: NeRD motion → nonlearned contact-force reconstruction → Newton rigid-body update. No MPM solve at inference.

Fix parameters before final evaluation; use separate episodes if calibration is needed. The first evaluation did not improve force accuracy; results follow.

ACCURACY DID NOT IMPROVE

The DEM prior increased force prediction loss

52 matched frames from two evaluation episodes. Parameters selected on two different episodes.
Force MSE averages squared error across frames, six bodies, and three force components.

METHOD	FORCE MSE [N²]	BOX FORCE RMSE [N]	PARTICLE BALANCE RESIDUAL	CONVERGED FRAMES
Original baseline	5,290	214.8	0.04%	0 / 52
Zero prior · new basis	6,877	225.2	14.78%	52 / 52
DEM · velocity friction	6,554	223.9	13.86%	52 / 52
DEM · contact history	6,524	224.0	13.92%	52 / 52

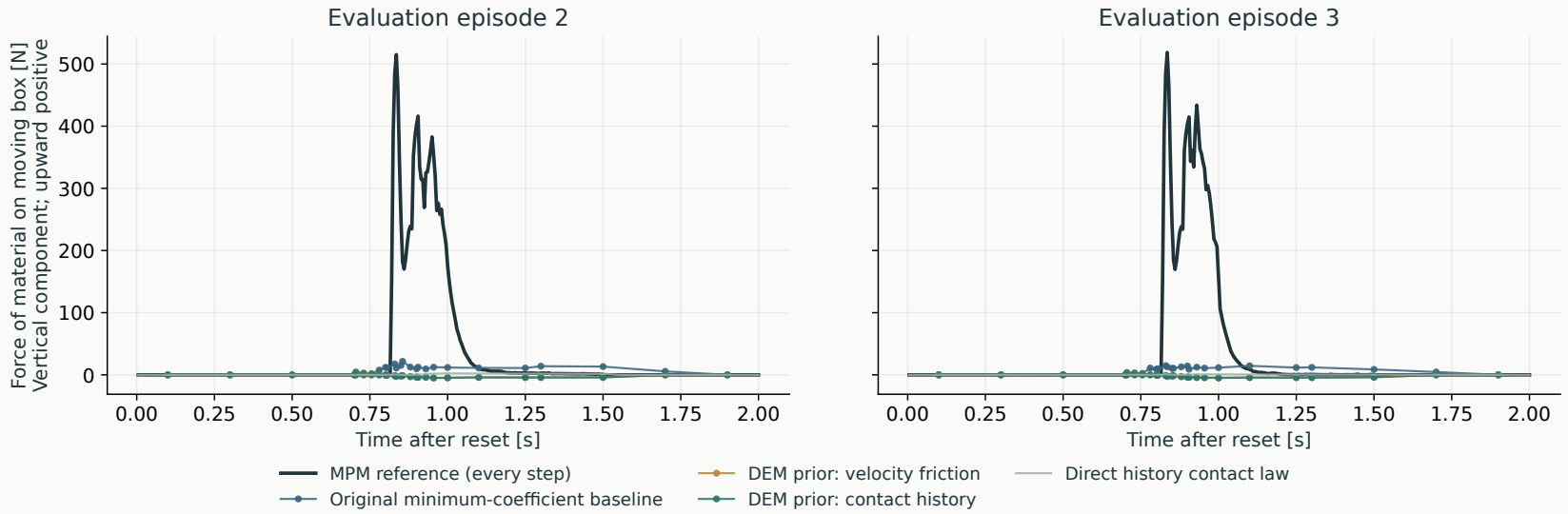
Contact history: 23.3% higher force MSE than the original baseline. The prior improves on the zero-prior variant with the same new force basis, but neither DEM variant meets the particle-balance criterion.

Both selected variants: stiffness 100 N/m; prior weight 0.1. No calibration candidate met both convergence and mean balance residual ≤ 5%.

Box RMSE uses the force-vector error; balance residual is normalized per frame by the particle inertial load after subtracting gravity.
The original baseline reached its iteration limit on all 52 frames. The new solver met its projected-gradient criterion. These are different convergence tests.
Data: DEM experiment (local file) · matched original baseline (local file) · summary (local file).

CONTACT MODEL REMAINS INSUFFICIENT

The reconstructed box force still misses the impact load



Material force on the moving box. Markers show evaluated frames; connecting lines interpolate between samples.

515–519 N
MPM force at each reference peak

−2.5 N
DEM history estimate at those same frames

0.08 / 0.39 N
Force change RMSE for 0.1 / 1 mm position offsets

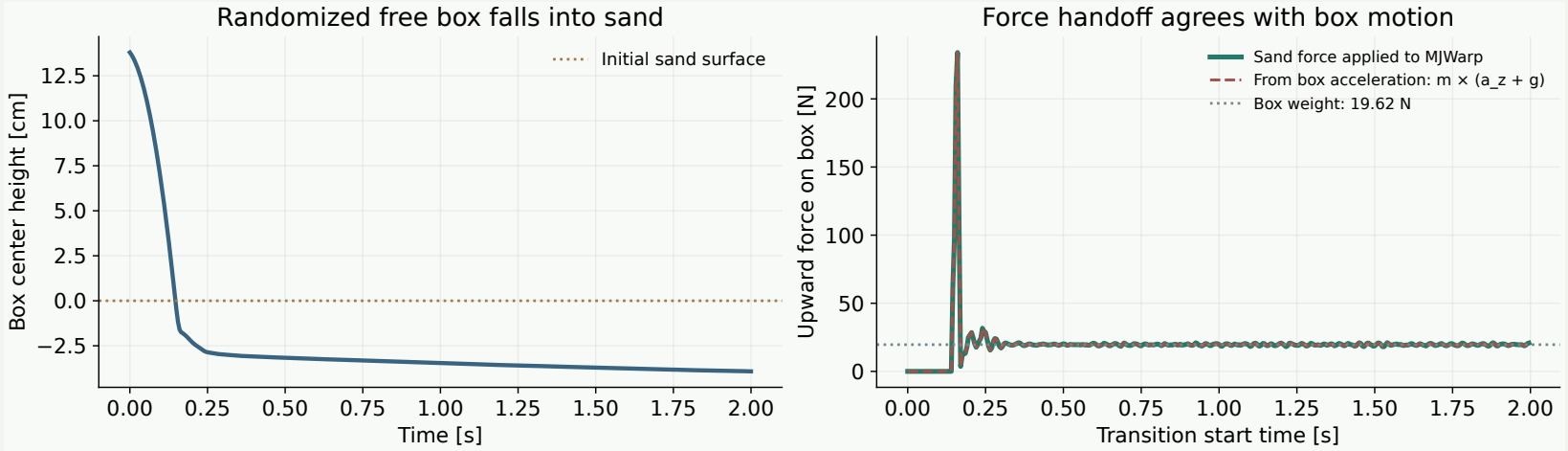
Conclusion: adding this effective contact law did not resolve the missing body loads. Revisit effective contact geometry and material relationships before using the estimate for two-way coupling.

Position sensitivity: 12 matched frames, all body-force components, persistent Gaussian particle offsets per coordinate. Velocities and accelerations unchanged.
Small sensitivity does not establish force accuracy. This experiment uses recorded MPM motion; no MPM solve is retained at inference.

MJWARP + MPM

The box now moves under gravity and contact

Randomize the 2 kg box's initial position, orientation, and velocity. Apply no commands after reset.
MJWarp advances the box; MPM advances the sand; the coupler exchanges motion and reactions.



Four episodes, 1,600 transitions: mean box force-balance residual 0.000002 N.
This validates reference coupling. Reconstructed-force accuracy is evaluated separately.

Box balance: mass × velocity change = gravity impulse + applied sand impulse + direct rigid-contact impulse.
Lagged coupling applies the previous MPM reaction; compare matching intervals. All runs use 25 MPM iterations maximum and a 5 ms timestep.
Latest experiment progress · MPM remains absent from the proposed NeRD inference loop.

NEW APPROXIMATION

Recover box reactions from contact motion

1. Free box motion

Advance current box velocity under gravity and known loads.

→

2. Material motion

Read the predicted next velocities of particles near the box.

→

3. Contact solve

Adjust box velocity to reduce closing motion at unilateral contacts.

→

4. Body reaction

Convert the box's momentum correction into contact force and torque.

```
contact impulse = rigid-body mass matrix × (corrected velocity − free velocity)
```

Additional physical information

Box mass, inertia, and contact velocity conditions. Measured future box acceleration is not an input.

Explicit approximation

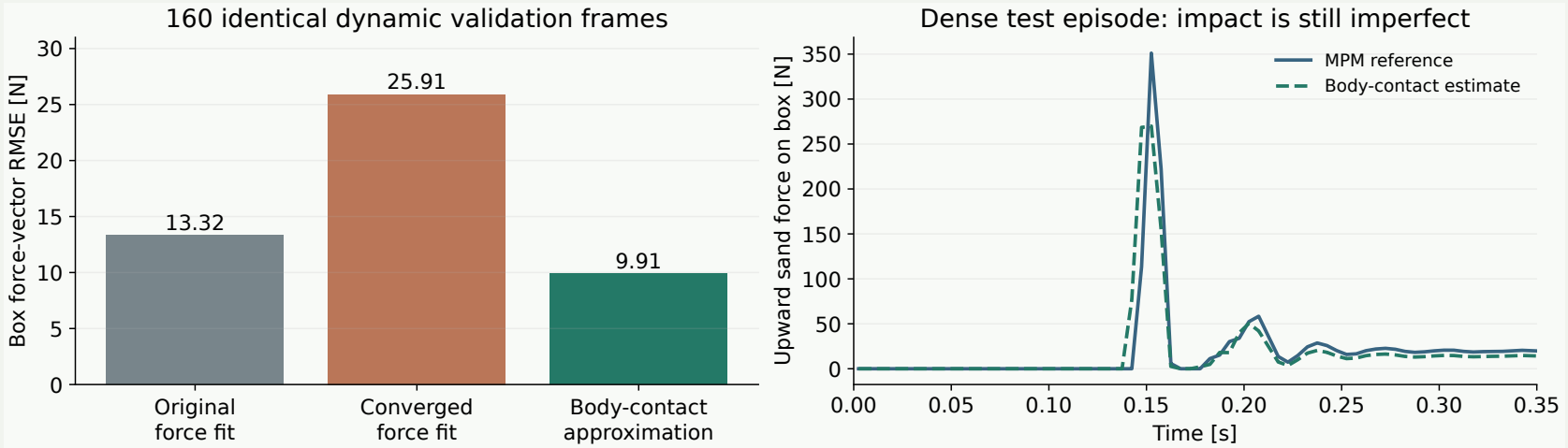
Predicted particle velocities define the material surface motion. An effective contact mass allows compliant relative motion.

Scope: dynamic-box reaction only. Full particle momentum consistency and separate fixed-body reactions remain unresolved.
Reference particle velocities stand in for NeRD predictions in these experiments. No MPM solver or force network runs inside the estimator.
The first variant uses normal contact only; a separate projected-contact iteration adds Coulomb friction.

MATCHED-FRAME COMPARISON

Box-force error decreased; impact and support errors remain

On 160 identical dynamic frames, box-force MSE decreased from 59.15 to 32.71 N²: **44.7% lower**.
The new parameters were selected using different calibration episodes.



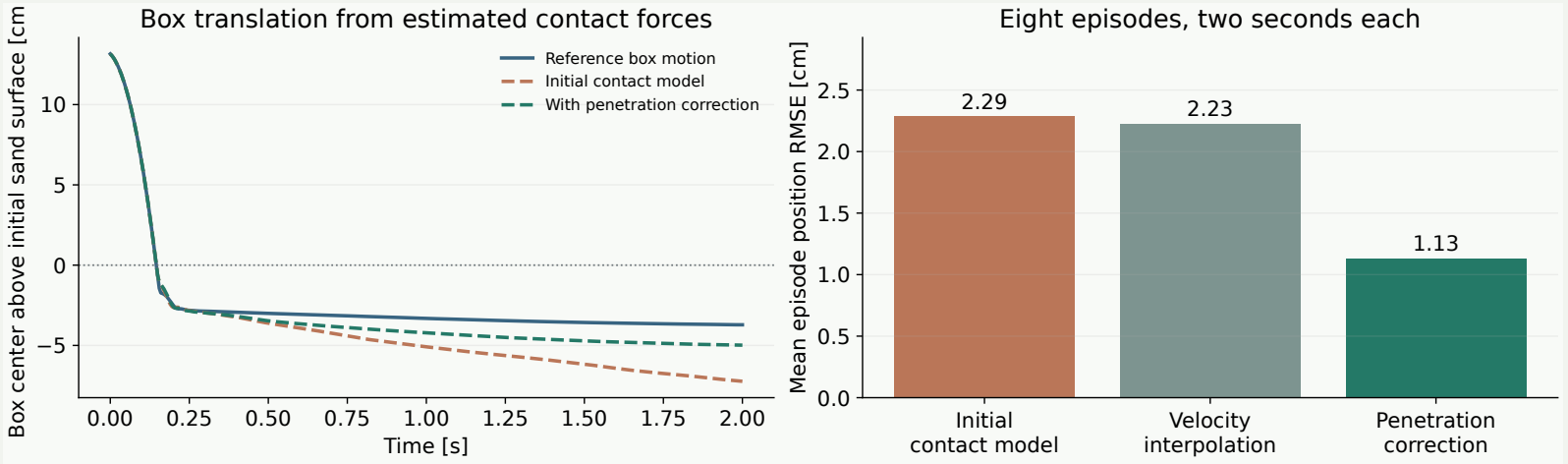
Separate dense test: 8 episodes, 3,200 transitions. Force-vector RMSE 10.69 N; relative force error 39.4%.
Steady upward force is approximately 12–15 N; the reference is 19.62 N.

Normal-contact fit: zero contact offset, effective contact mass = 10 × particle mass, start-of-step geometry. Median CPU time ≈ 1.1 ms for one box.
Coulomb-friction variant: test force RMSE 14.46 N, torque RMSE 0.320 N m; normal-only torque RMSE 0.398 N m.
Earlier inverse models estimate all six bodies; the new estimator handles the dynamic box only. Latest progress

OFFLINE REPLAY

Contact correction improves box translation

Start from the recorded initial box state. Estimate its contact force at each step, then integrate its motion.
Particle motion remains the recorded reference; no MPM solver runs during this replay.



Mean position RMSE: 2.29 → 1.13 cm. Mean orientation RMSE: 9.57 → 8.93°.
The revised contact model also lowers force-vector RMSE from 10.69 to 9.85 N on the same 3,200 steps.

Correction: estimate particle velocity at the box surface and add a separating velocity proportional to penetration depth.
Eight 2-second episodes. Reference-force replay gives 1.34 µm mean position RMSE, checking force timing and integration.
Recorded particles cannot respond to box-motion errors. A full coupled NeRD rollout remains untested.

RESEARCH STATUS

The hypothesis is clear; the particle reconstruction is open

Established by the sources

- Cable NeRD separates learned motion from analytical reactions.
- Newton exposes collider identity and body-reaction coupling.
- L-GBND represents terrain–tool interactions with typed nodes.

To establish experimentally

- A suitable material law and reconstruction discretization.
- Sufficient particle state for stable inference.
- Accurate reactions from imperfect learned motion.
- A net speed benefit for the coupled simulation.

[1] Cable NeRD · slides 23–24

[2] L-GBND · Liu, Li & Hauser

[3] L-GBND · released architecture

[4] XPBD · Macklin et al.

[5] Newton · coupled example

[6] Newton · implicit MPM solver